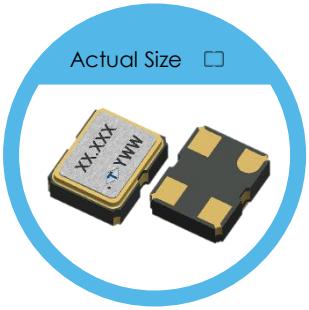
TAITIEN
Our Performance, Your Reputation

OY Type

2.5 x 2.0 mm SMD Ultra Low Power SMD CMOS Output Crystal Oscillator



RoHS Compliant

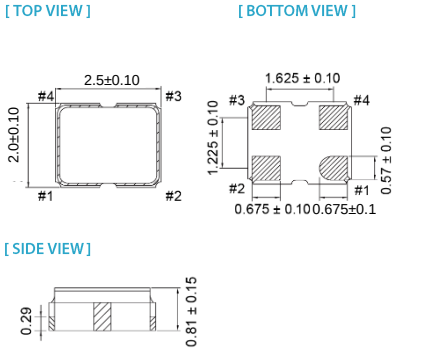
FEATURES

- Ultra Low Power Supply Voltage: 0.9V, 1.2V, 1.5V Supply Options
- Singled-end Output: CMOS
- Frequency Support from 1MHz to 50MHz
- Low Noise Typical: 0.3ps at 12kHz to 20MHz Frequency Offsets
- Temperature Range: -40 to 85°C Operation

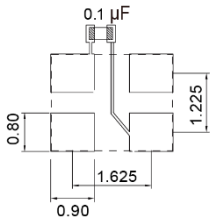
TYPICAL APPLICATION

- IoT, Smartphone, Digital Camera
- Game Console, Wearable Device, Digital Consumer Electronics

DIMENSION (mm)



SOLDER PAD LAYOUT (mm)



To ensure optimal oscillator performance, place a by-pass capacitor of 0.1µF as close to the part as possible between V_{DD} and GND pads.

PIN FUNCTION (mm)

PIN#	FUNCTION
1	Tri-State
2	GND
3	Output
4	V_{DD}

ELECTRICAL SPECIFICATION

Parameter			0.9V		1.2V		1.5V		Unit	Test Condition
			Min.	Max.	Min.	Max.	Min.	Max.		
Supply Voltage Variation (V_{DD})			V_{DD} – 5%	V_{DD} + 5%	V_{DD} – 5%	V_{DD} + 5%	V_{DD} – 5%	V_{DD} + 5%	V	
Frequency Range			1	50	1	50	1	50	MHz	
Supply Current	At 15pF Load			1.5		2		3	mA	
	No Load Condition	1 MHz ≦ Fo < 10 MHz		0.9		1		1.2	mA	
		10 MHz ≦ Fo < 20 MHz		1		1.2		1.2	mA	
		20 MHz ≦ Fo < 50 MHz		1.2		1.5		1.5	mA	
Duty Cycle			45	55	45	55	45	55	%	
Output Level	Output High		90% V_{DD}		90% V_{DD}		90% V_{DD}		V	
	Output Low			10% V_{DD}		10% V_{DD}		10% V_{DD}	V	
Transition Time: Rise/Fall Time	1 MHz ≦ Fo < 50 MHz			6		6		6	nSec	Transition times are measured between 10% and 90% of V_{DD}, with an output load of 15 pF.
Startup Time				5		5		5	mSec	
Tri-State (Input to Pin 1)	Enable (High Voltage or Floating)		0.7 x V_{DD}		0.7 x V_{DD}		0.7 x V_{DD}		V	
	Disable (Low Voltage or GND)			0.3 x V_{DD}		0.3 x V_{DD}		0.3 x V_{DD}	V	
Output Loading			15		15		15		pF	
Standby Current				100		100		100	µA	
Aging (@25°C, 1st year)				±3		±3		±3	ppm	
Storage Temp. Range			-55	+125	-55	+125	-55	+125	°C	
Phase Noise	At V_{DD}=1.2V, Fout=24MHz	1kHz offset	-130		-133		-135		dBc/Hz	
		10kHz offset	-140		-143		-143		dBc/Hz	
		100kHz offset	-148		-150		-150		dBc/Hz	
		1MHz offset	-152		-155		-155		dBc/Hz	
Period Jitter (pk-pk)			-	40	-	40	-	40	pSec	
RMS Phase Jitter (Integrated 12 kHz~20 MHz)			-	1	-	1	-	1	pSec	

FREQ. STABILITY vs. TEMP. RANGE

ppm / Temp.(°C)	±25	±50
-10 ~ +60	○	○
-20 ~ +70	○	○
-40 ~ +85	△	○

○: Available △:Conditional x: Not Available
Inclusive of calibration @ 25°C ,operating temperature rrange,input Voltage variation,load variation,aging (1st year),shock,and vibration

Note: not all combination of options are available. Other specifications may be available upon request.
Specifications subject to change without notice.

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